

Data Driven Modeling of monoclonal antibody(mAb) production in bioreactor using LSTM (Long Short-Term Memory) Neural Networks



Course Code : CLD411

Course Name : B.Tech Project -1

Department of Chemical Engineering , IIT Delhi

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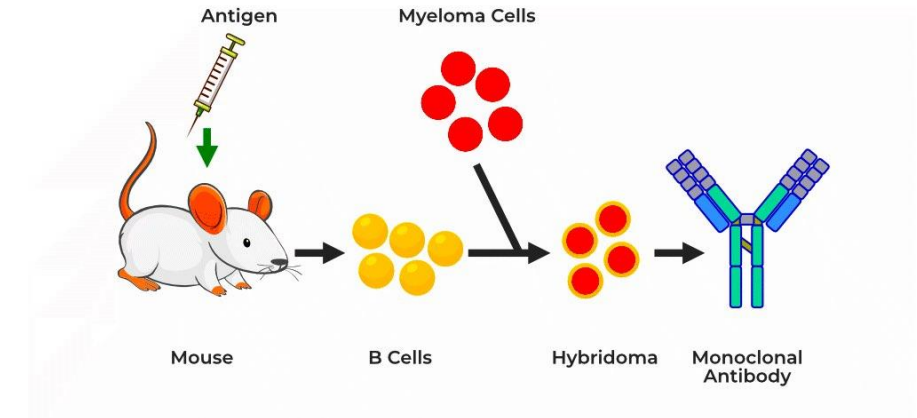
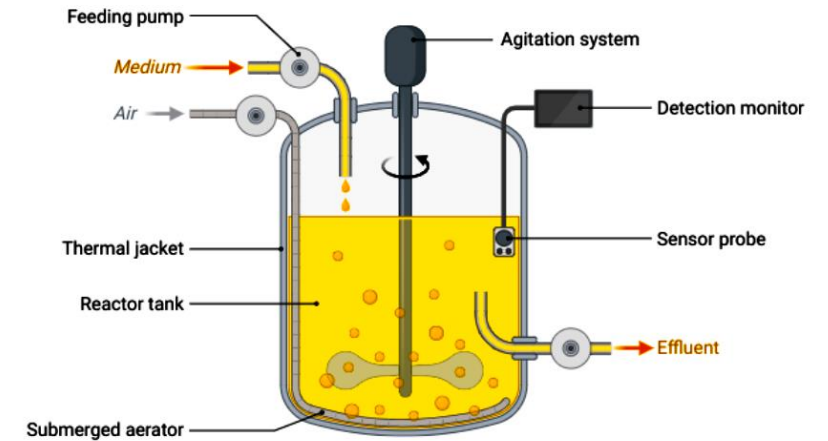
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Introduction

- ❖ Monoclonal antibodies(mAbs) are highly specific lab-engineered proteins designed for treatments for diseases.
- ❖ The production of mAbs is typically done using Chinese Hamster Ovary (CHO) cells.
- ❖ Bioreactors are vessels that support biologically active environments to cultivate cells.



Problem Statement & Objectives :

Accurately predicting mAb concentrations during manufacturing is critical but challenging due to:

To address these challenges, we aim to:

- **Complex Dynamics:**

- Bioproduction processes involve non-linear reactions influenced by variables like pH, temperature, and nutrient levels.

- **Real-Time Monitoring Needs:**

- Current methods may require extensive time and resources for sampling and analysis, slowing down optimization and production.

- **Data Scarcity:**

- Limited availability of high-resolution, time-series data complicates traditional modeling approaches.

- Build a **predictive model**

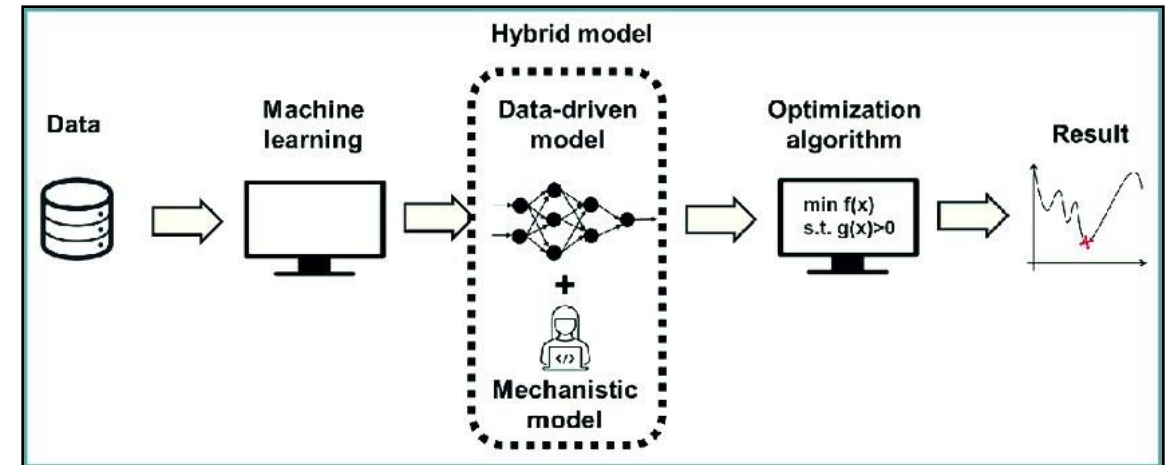
- using **LSTM neural networks**, a deep learning architecture well-suited for sequential data.

- Enable accurate forecasting of mAb concentration over time with:

- Reduced dependence on exhaustive experimental setups.
- The ability to optimize and monitor processes in real-time.

- Support scaling and

- refinement of bioproduction, ensuring consistency in mAb quality and yield.



Model Development of Bioreactor :

These are the differential equations that are governing the dynamics of bioreactor:

$$\frac{dV}{dt} = Q_i - Q_{o,s} - Q_{o,r} \quad (1)$$

$$\frac{dX_v V}{dt} = (\mu - \mu_{death}) X_v V - Q_{o,s} X_v \quad (2)$$

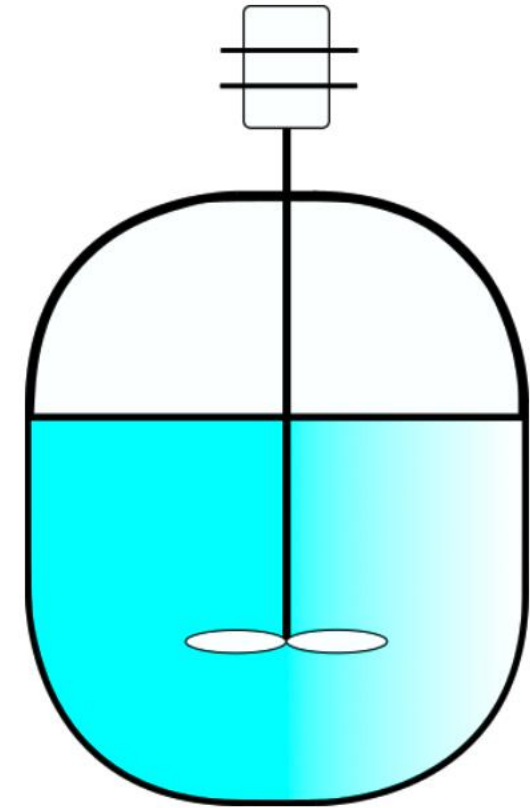
$$\mu = \mu_{max} f_{lim} f_{inh} f_{temp} \quad (3)$$

$$\mu_{death} = \mu_{death,max} f_{death,temp} \quad (4)$$

$$\frac{dX_v V}{dt} = \left[\mu_{max} f_{lim} f_{inh} \exp\left(-\frac{K_1}{T}\right) - \mu_{death,max} \exp\left(-\frac{K_2}{T}\right) \right] X_v V - Q_{o,s} X_v \quad (5)$$

$$f_{lim} = \frac{[Glc]}{K_{Glc}(X_v) + [Glc]}; K_{Glc}(X_v) = K_{Glc} X_v \quad (6)$$

$$f_{inh} = \frac{KI_{Lac}}{KI_{Lac} + [Lac]} (1 - KI_{mAb}[mAb]) \quad (7)$$



Scheme of a batch bioreactor

$$\frac{dX_d V}{dt} = \left[\mu_{death,max} \exp\left(-\frac{K_2}{T}\right) \right] X_v V - Q_{o,s} X_d \quad (8)$$

$$Viability = \frac{X_v}{X_v + X_d} \times 100\% \quad (9)$$

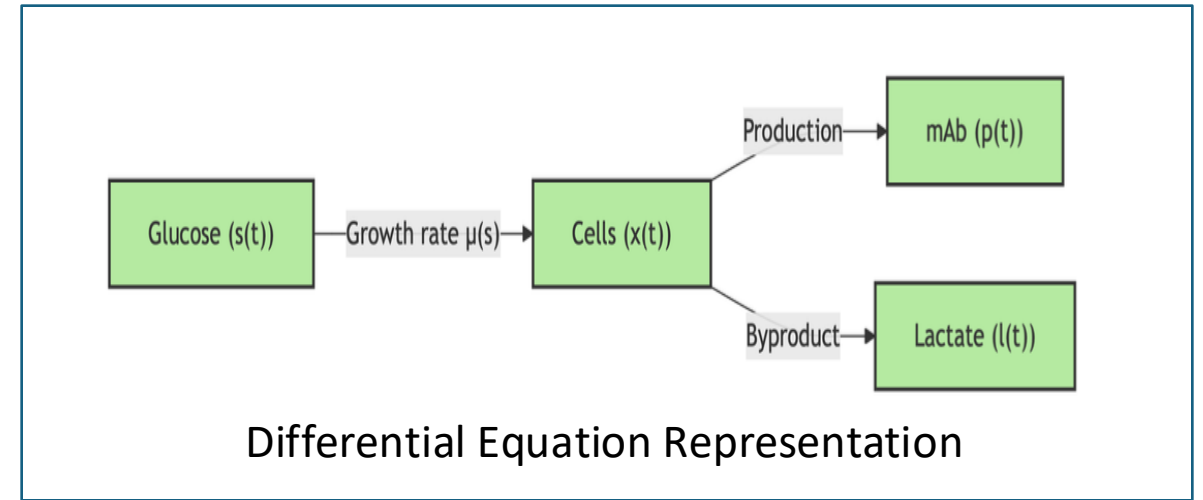
$$\begin{aligned} \frac{d[Metabolite]V}{dt} = & Q_i [Metabolite_{inlet}] - (Q_{o,s} + Q_{o,r}) [Metabolite] \\ & + q_{Metabolite} X_v V \end{aligned}$$

$$q_{Glc} = -\frac{\mu}{Y_{X/Glc}} - m_{Glc} \quad (11)$$

$$q_{Lac} = \left(\frac{\mu}{Y_{X/Lac}} + Y_{Lac/Glc} q_{Glc} \right) \frac{Lac_{max1} - [Lac]}{Lac_{max1}} + m_{Lac} \frac{Lac_{max2} - [Lac]}{Lac_{max2}} \quad (12)$$

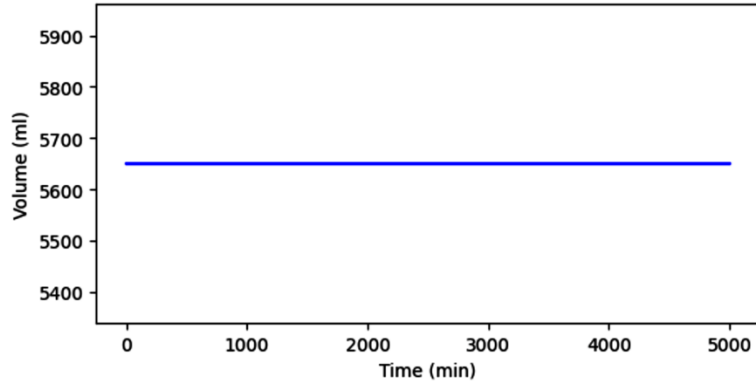
mAb production was calculated with the material balance, by Eq. (13)

$$\frac{d[mAb]V}{dt} = -Q_{o,s} [mAb] + q_{mAb} X_v V \quad (13)$$

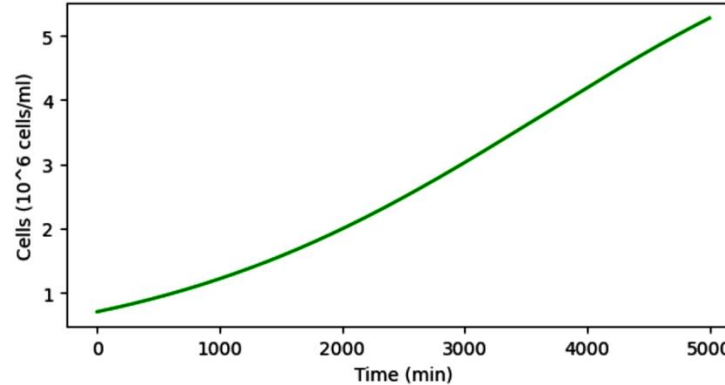


Simulation of Bioreactor Variables :

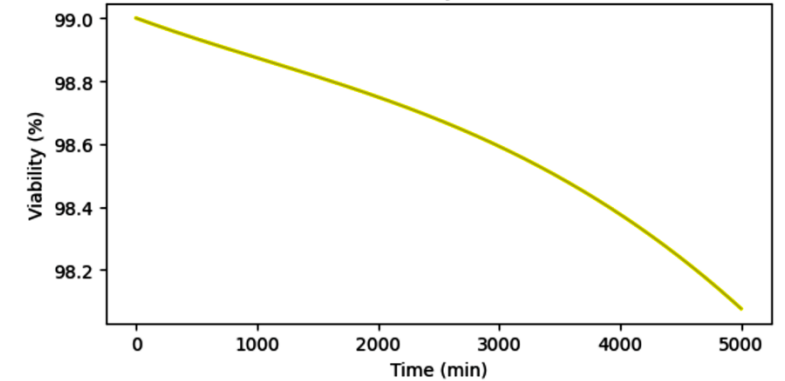
Volume of Mixture (v)



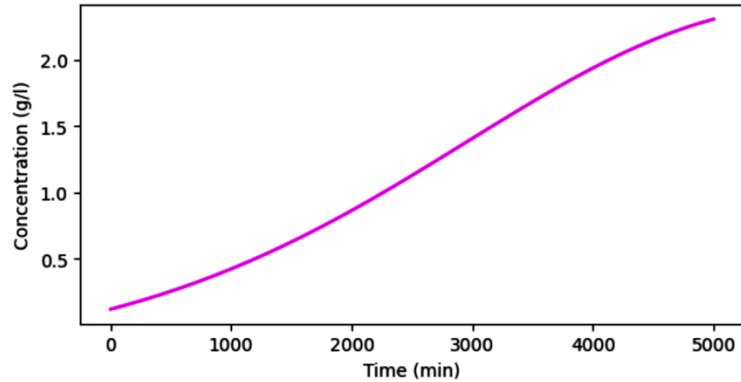
Viable Cells (x)



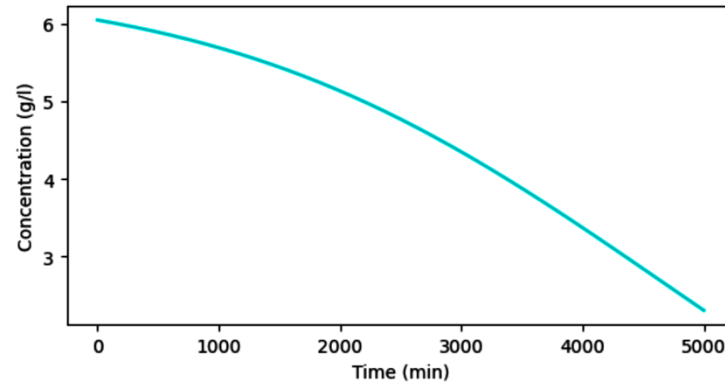
Viability (vb)



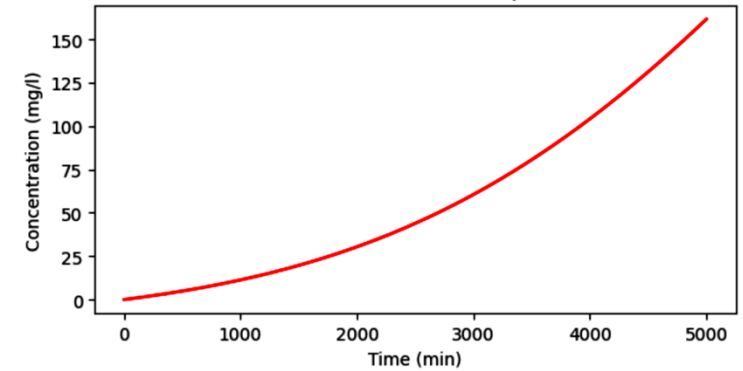
Lactate Concentration (l)



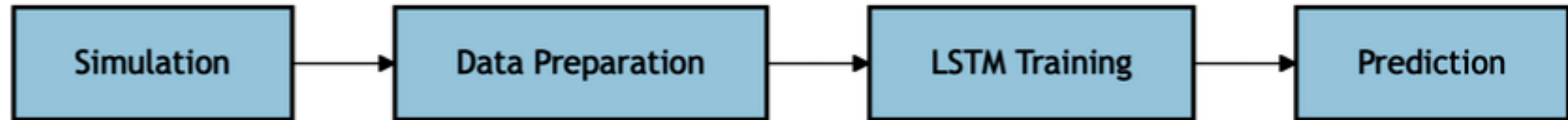
Glucose Concentration (s)



mAb Concentration (p)

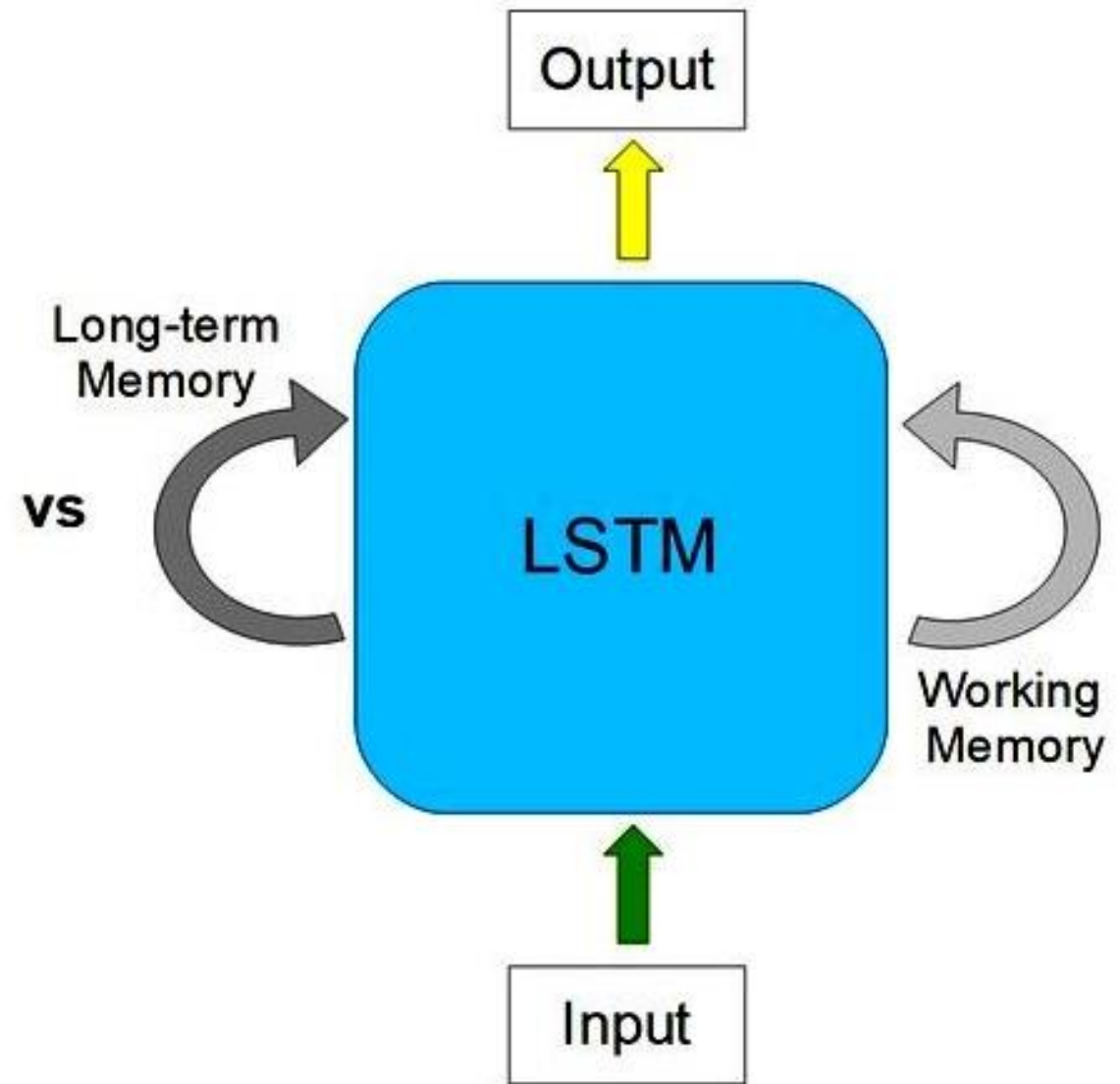


Experimental Workflow from Simulations to Prediction



Why Data Driven Modeling by LSTM Neural Network :

- **Time-Series Data:** LSTM networks are designed to process sequential data, our process involve time-dependent variables.
- **Long-Term Memory:** This mechanism allows the network to retain information from previous time steps, which is crucial for understanding the cumulative effects of past events on the current state of the bioreactor.
- **Automatic Feature Extraction:** LSTM networks can automatically learn relevant features from the input data, reducing the need for manual feature engineering.
- **Forecasting:** LSTM networks can be used to predict future states of the bioreactor based on past observations.



Model Architecture of LSTM



Structure:

- Input Layer:** Time-series data reshaped to (None, 20, 1).
- LSTM Layer 1:** Processes sequential dependencies; output (None, 20, 50).
- Dropout 1:** Rate of 0.2 to reduce overfitting.
- LSTM Layer 2:** Extracts high-level patterns; output (None, 50).
- Dropout 2:** Another 0.2 rate for robustness.
- Dense Output Layer:** Final prediction of mAb concentration (None, 1).



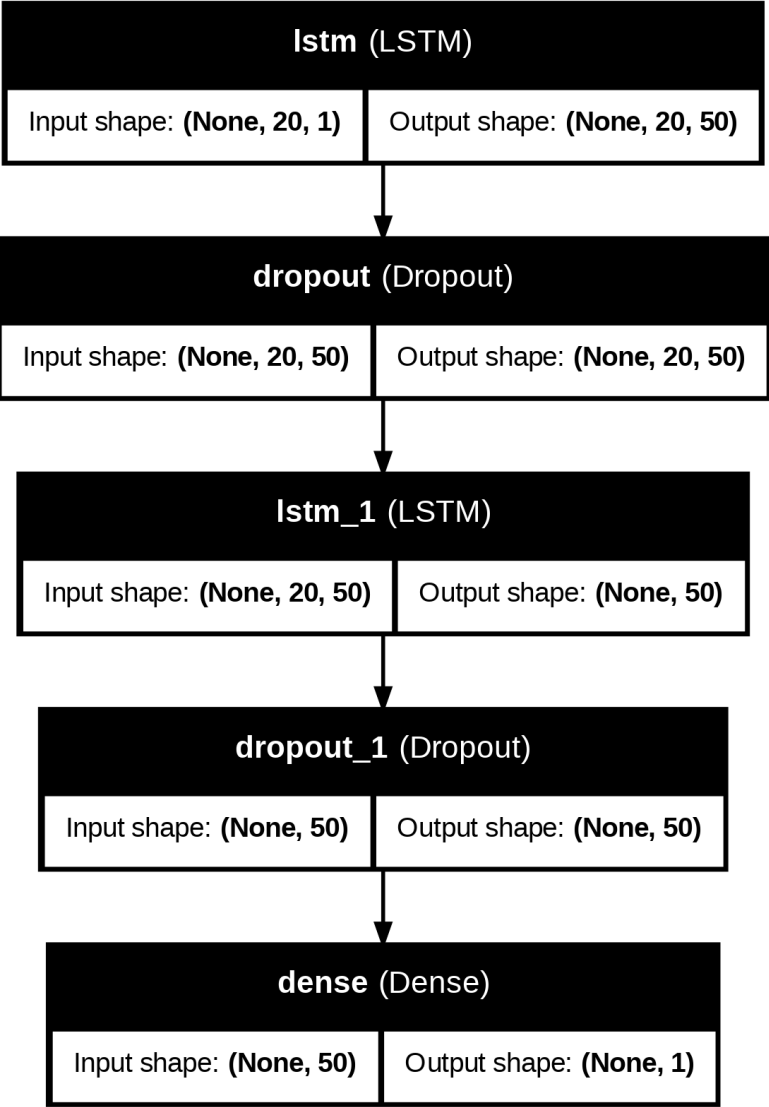
Key Features:

- LSTM Layers:** Capture temporal patterns in sequential data.
- Dropout Layers:** Improve generalization by preventing overfitting.
- Dense Layer:** Transforms learned features into accurate predictions.



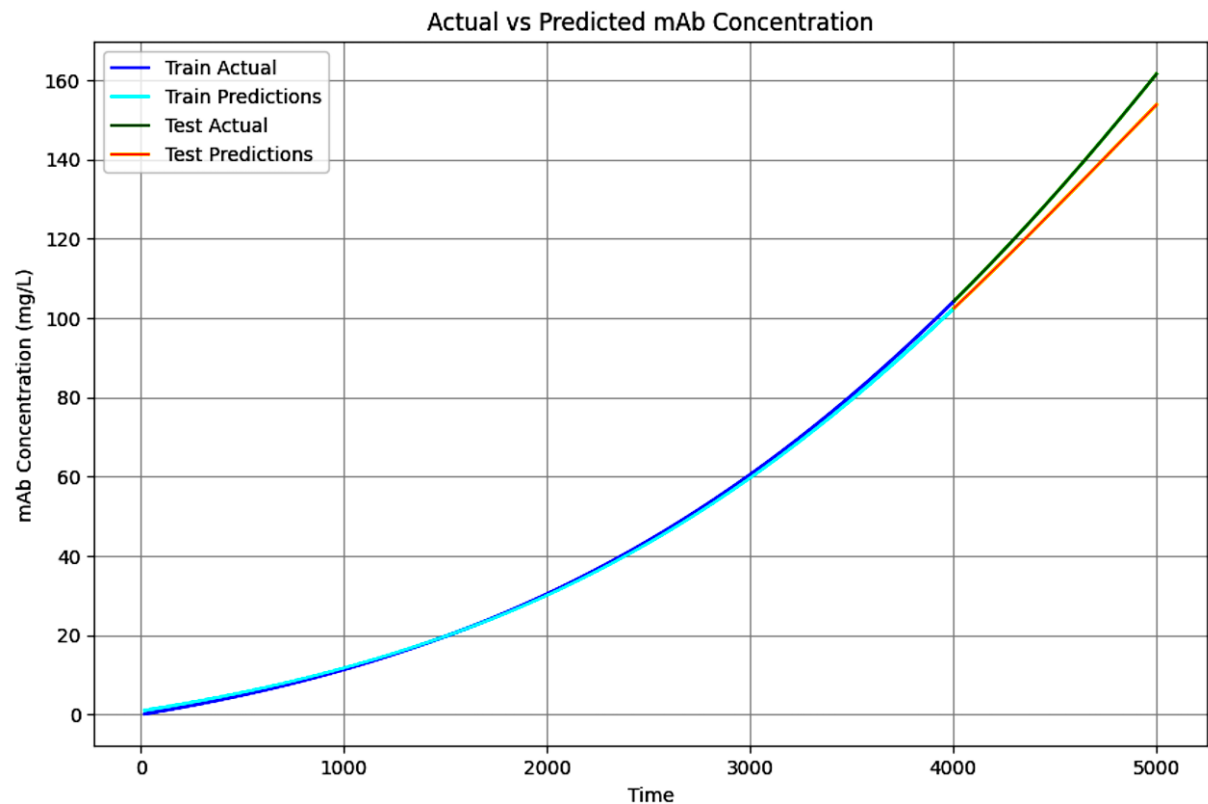
Why This Architecture?

- Models the non-linear dynamics of mAb concentration effectively.
- Handles time-dependent variations, ensuring precise forecasts.

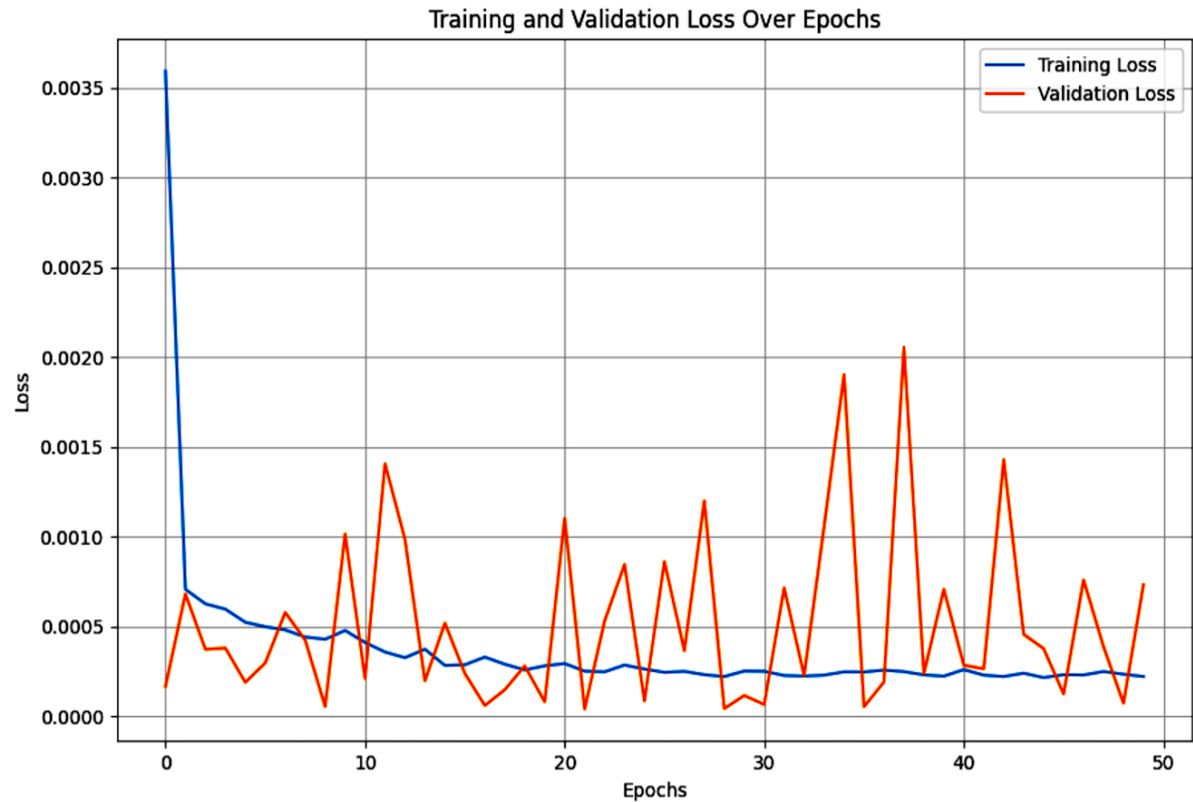


Architecture

Results :

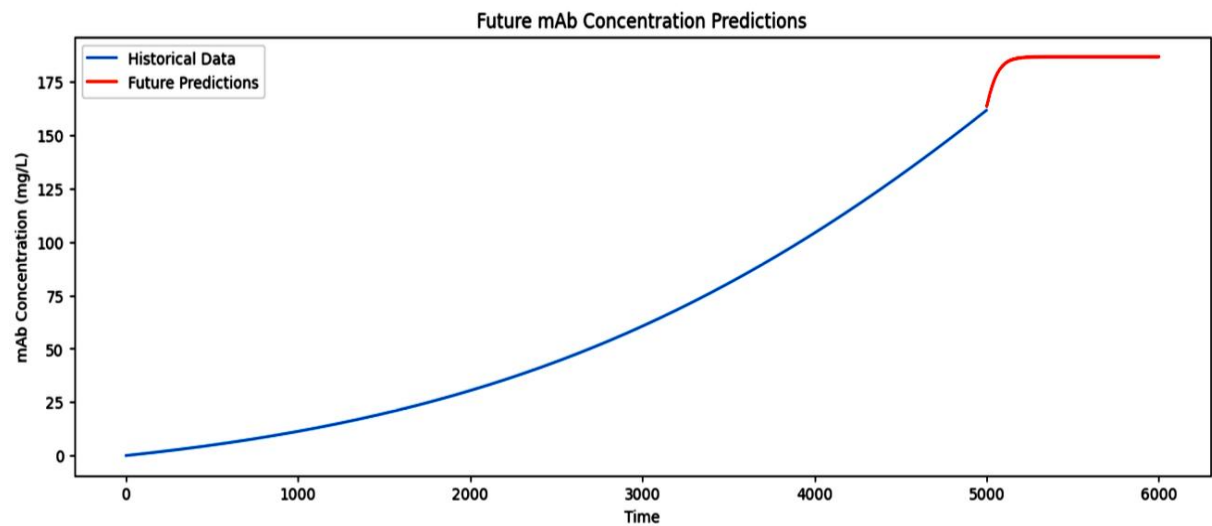


Training and Validation Performance (Loss Curve)

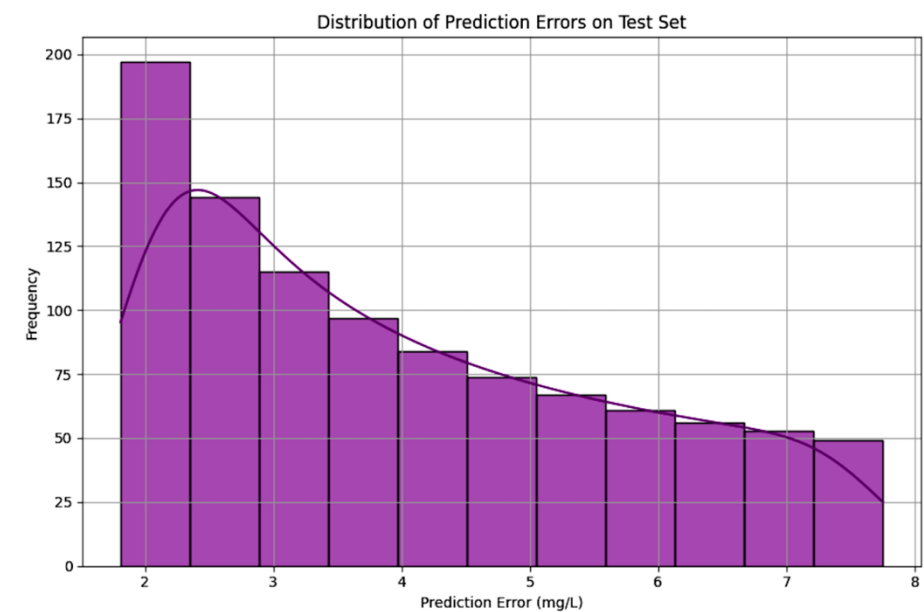


Actual vs Predicted (Train & Test)

Results :



Future Forecast of mAb concentration



Error Distribution of Test Set

Metric	Value
Mean Squared Error (MSE)	0.0124
Mean Absolute Error (MAE)	0.0852
Root Mean Squared Error (RMSE)	0.1114

Error Metrics (MSE, MAE, RMSE)

References :

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